Explicit Strong-Coupling Bounds and Vacuum-Adapted Coercivity for SU(N) Lattice Gauge Hamiltonians

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October 28, 2026 Version 1.0

Abstract

We give quantitative spectral estimates for finite-lattice SU(N) gauge Hamiltonians and identify the additional estimates needed for a continuum mass-gap argument. A sector-valued Kirkwood–Thomas expansion extends a weak-interaction argument to bounded interactions on at most s sites, with constants independent of the local Hilbert-space dimension. For a specified normalization of the Kogut–Susskind Hamiltonian in three spatial dimensions, this yields a finite-volume gap $\Delta_{\mathrm{lat}} \geq 3g^2/16$ whenever $g^2 \geq 59.9$, uniformly in $N \geq 2$ and lattice size. We also give a counterexample to an intermediate counting assertion in the original Bravyi–DiVincenzo–Loss preprint and show that its conservative numerical constant survives the correction. For a fixed block in the actual interacting finite-lattice vacuum, a Feynman–Kac comparison yields a positive discarded-sector bound uniform in the exterior volume. Finally, an exact Schur-complement comparison adds inverse-gap costs when the induced coarse inner product is retained. The first two spectral estimates are unconditional within their stated finite-lattice hypotheses; their iteration into a nonlinear multiscale construction is not established. We do not prove continuum Yang–Mills existence, a continuum mass gap, or confinement. The accompanying exact arithmetic checks certify specified constants and counterexamples, not the full analytic arguments.

1 Introduction and scope

The Yang–Mills existence and mass-gap problem asks for a nontrivial four-dimensional quantum field theory for every compact simple gauge group, together with a strictly positive physical mass gap and the required field-theoretic axioms [5]. A positive eigenvalue gap for one finite regulator is a much weaker assertion. Even a lattice-size uniform gap at fixed strong coupling does not by itself control the weak bare-coupling trajectory used in a continuum limit.

This paper isolates several finite-regulator results and an exact conditional comparison. The principal quantitative result is a dimension-independent extension of the creation-operator estimates of Bravyi, DiVincenzo and Loss [1]. The extension is applied to a Hamiltonian lattice gauge model of Kogut–Susskind type [3]. Weak-interaction stability of gapped product systems, including systems with infinite-dimensional local spaces, is already established in Yarotsky's work [2]; the existence of a sufficiently strong-coupling lattice gap is not claimed as a new phenomenon here. Our emphasis is the explicit sector estimate, the resulting sufficient constants, and careful separation of finite-lattice and continuum claims. We make no claim that these constants are optimal or that their precise form has not appeared elsewhere.

The second result concerns the actual ground-state measure, rather than a product reference vacuum. A local Feynman–Kac comparison controls the oscillation of the ground function over a fixed block independently of the size of its exterior. Product-Haar Poincaré comparison then gives coercivity on the sector discarded by conditional expectation. The heat-kernel ingredient is the compact nonnegative-Ricci case of the Li–Yau inequality [4]. This result holds at every positive coupling for a fixed finite block, but its constants do not remain uniform when that block becomes an extensive elimination layer.

The final comparison records which operator must be controlled after exact elimination. A Schur form alone omits the norm of its graph lift. Keeping that norm gives a harmonic gap bound and additive inverse-gap costs. We provide counterexamples to two tempting replacements of this argument: ignoring the graph norm, and identifying quantum Schur elimination with a transformation that preserves quadratic oscillator Hamiltonians.

1.1 What is proved, and what is conditional

Result	Proved scope	Limitation
Explicit strong-coupling stability	Finite systems; constants uniform in size and on-site dimension	No explicit-threshold infinite-volume construction or zero-spacing limit
Gauge-Hamiltonian corollary	Specified $\mathrm{SU}(N)$ normalization; all $N \geq 2$ at fixed g^2	Uniformity is not at fixed 't Hooft coupling
Actual-vacuum block coercivity	Fixed finite block; all exterior volumes and configurations	No uniform whole-layer or iterated coarse-operator estimate
Schur gap transfer	Exact operators and stated form domains, with their induced metrics	Application to successive Yang–Mills scales remains conditional

1.2 Conventions and provenance

Let E be a finite set of oriented lattice links, with one group variable per link, and let $\mathcal{P}$ be its plaquettes. Haar measure is normalized. The kinematic Hilbert space and dimensionless Hamiltonian are

$$\mathcal{H}_E = L^2(\mathrm{SU}(N)^E, \mathrm{d}U), \qquad x = g^2 > 0, \quad N \geq 2, \tag{1.1}$$

$$H_E(x) = \frac{x}{2} \sum_{\ell \in E} C_\ell + \frac{b_N}{x} \sum_{p \in \mathcal{P}} (1 - T_p), \qquad T_p = \frac{1}{N} \operatorname{Re} \operatorname{Tr} U_p. \tag{1.2}$$

The nonnegative Laplace–Beltrami operator $C_\ell = -\Delta_\ell$ uses the Casimir normalization

$$C_F = \frac{N^2 - 1}{2N}, \qquad \inf\bigl(\operatorname{spec} C_\ell \setminus \{0\}\bigr) = C_F. \tag{1.3}$$

The magnetic coefficient is an explicitly declared parameter. The uniform strong-coupling statement assumes $0 \leq b_N \leq 2N$; the finite-block theorem allows any fixed $b = b_N \geq 0$. Changing conventions requires transforming this parameter and the energy normalization. If $a > 0$ is the lattice spacing, physical excitation energies in units $\hbar = c = 1$ are eigenvalues of $(H_E - E_0)/a$. The physical gauge-invariant space is the invariant subspace under the local $\mathrm{SU}(N)$ gauge transformations. Both the kinematic space and this restriction are specified whenever a gap is asserted.

The arguments assembled here were developed in September 2026 research notes. The earlier spectral paper [6] contains the fixed-spacing qualitative strong-coupling application. The later quantitative and vacuum-adapted notes are included in the October 5, 2026 versioned repository

archive [7]. The present paper gives a standalone exposition, exact supporting certificates, and an explicit account of what is still missing. These dates document provenance; they are not a claim of priority over other research.

2 An explicit finite-volume strong-coupling bound

This section gives a quantitative version of stability of a gapped product Hamiltonian under bounded local perturbations. The argument follows the Kirkwood–Thomas method used by Bravyi, DiVincenzo, and Loss [1], with Hilbert-space norms on excitation sectors in place of absolute values of qubit coefficients. The existence of a sufficiently small perturbative window, including for infinite-dimensional site spaces, is established in the literature [2]. The purpose here is to track constants for the specified Hamiltonian lattice gauge model. Neither the method nor the existence of strong-coupling lattice gaps is claimed to be new. The derivation and associated records originated in the research archive [7]; the earlier spectral-bounds paper [6] gives the corresponding non-explicit stability application.

2.1 A finite-system stability criterion

Let $\mathcal{L}$ be a nonempty finite set of sites. At each site u, let $\mathcal{H}_u$ be a separable Hilbert space and let h_u be a nonnegative self-adjoint operator with compact resolvent. Assume that h_u has a normalized simple zero eigenvector Ω_u and that, for a common $\Delta > 0$,

$$h_u\big|_{\Omega_u^\perp} \geq \Delta. \tag{2.1}$$

The inequality is vacuous if $\mathcal{H}_u$ is one-dimensional. On $\mathcal{H} = \bigotimes_{u\in\mathcal{L}} \mathcal{H}_u$, set

$$H_0 = \sum_{u\in\mathcal{L}} h_u, \qquad V = \sum_{p\in\mathcal{P}} V_p, \qquad H(\varepsilon) = H_0 + \varepsilon V. \tag{2.2}$$

Here $\mathcal{P}$ is a finite indexed family of nonempty site sets, each of cardinality at most an integer $s \geq 1$. Each V_p is self-adjoint, supported on p, and has norm at most J. Every site belongs to at most D members of $\mathcal{P}$, counted with their multiplicities. We may take $D, J > 0$; when there is no perturbation the conclusion below is immediate. The coupling ε is real.

Define the dimensionless parameter and positive constants

$$\lambda = \frac{|\varepsilon| DJ}{\Delta}, \tag{2.3}$$

$$\gamma_0(s)^2 = \sum_{t=1}^{s} \frac{\binom{s-1}{t-1}}{t^2}, \qquad \gamma_1(s)^2 = (s+1)^2 \sum_{t=0}^{s} \frac{\binom{s}{t}}{(t+1)^2}, \tag{2.4}$$

$$\beta_0(s) = \gamma_0(s), \qquad \beta_k(s) = 2^k\big(\gamma_0(s)s^k + k\gamma_1(s)s^{k-1}\big) \quad (1 \leq k \leq 2s), \tag{2.5}$$

$$q_s(r) = \sum_{k=0}^{2s} \frac{\beta_k(s)}{k!} r^k. \tag{2.6}$$

Theorem 2.1 (Dimension-independent finite-volume stability)**.** *Suppose there is a number $r > 0$ such that*

$$\lambda q_s(r) \leq r, \qquad 2\lambda q_s'(r) < 1. \tag{2.7}$$

Then $H(\varepsilon)$ has a simple ground eigenvalue E_0 and

$$\operatorname{spec} H(\varepsilon) \subset \{E_0\} \cup [E_0 + \Delta/2, \infty). \tag{2.8}$$

The criterion depends on s, D, J, Δ and ε, but not on the number of sites, their Hilbert-space dimensions, or the upper onsite spectrum.

We first prove the theorem when all site spaces are finite-dimensional. The compact-resolvent extension is proved at the end of this subsection.

For $M \subset \mathcal{L}$, let $\mathcal{K}_M = \bigotimes_{u\in M} \Omega_u^{\perp}$ and $\Omega_M = \bigotimes_{u\in M} \Omega_u$. The full Hilbert space is the orthogonal sum of the sectors $\mathcal{K}_M \otimes \Omega_{\mathcal{L}\setminus M}$. We identify a sector with $\mathcal{K}_M$, and write Π_M for the corresponding projection followed by this identification. In particular,

$$\sum_{M\subset\mathcal{L}} \|\Pi_M \xi\|^2 = \|\xi\|^2. \tag{2.9}$$

Let $\Omega = \bigotimes_{u\in\mathcal{L}} \Omega_u$. For $c_M \in \mathcal{K}_M$, $M \neq \varnothing$, define

$$A[c_M] = |c_M\rangle\langle\Omega_M| \otimes I_{\mathcal{L}\setminus M}, \qquad C = \sum_{M\neq\varnothing} A[c_M], \qquad \|C\|_1 = \max_{u\in\mathcal{L}} \sum_{M\ni u} \|c_M\|. \tag{2.10}$$

The decomposition of C is unique, as is seen by applying it to Ω. We call C a creation operator. The coefficients c_M are vectors, including when the excitations within a sector are entangled.

Two such elementary operators commute. If their supports intersect, their product in either order is zero: a vacuum bra on an overlapping site annihilates an excited factor. If their supports are disjoint, they commute by tensor factorization. Consequently all creation operators commute, and every product of more than $|\mathcal{L}|$ elementary creation operators vanishes. Their exponentials are therefore finite polynomials. Also, $\|A[c_M]\| = \|c_M\|$.

Put $H_0^M = \sum_{u\in M} h_u$ on $\mathcal{K}_M$. Then

$$H_0^M \geq \Delta|M|, \qquad [H_0, A[c_M]] = A[H_0^M c_M]. \tag{2.11}$$

Thus a second commutator of H_0 with creation operators is zero. We use $\mathrm{ad}_C(X) = [C, X]$.

Lemma 2.2 (Support and commutator depth)**.** *Fix $p \in \mathcal{P}$ and elementary creation operators $A[c_{M_1}], \ldots, A[c_{M_k}]$. Set $N_* = \bigcup_{i=1}^k M_i$ and*

$$X = \mathrm{ad}_{A[c_{M_1}]} \cdots \mathrm{ad}_{A[c_{M_k}]}(V_p).$$

If $X \neq 0$, each M_i intersects p. If $\Pi_M X \Omega \neq 0$, then

$$N_* \setminus p \subset M \subset N_* \cup p, \qquad M = (N_* \setminus p) \cup T, \quad T \subset p. \tag{2.12}$$

Moreover, for arbitrary creation operators $C_1, \ldots, C_k$,

$$\mathrm{ad}_{C_1} \cdots \mathrm{ad}_{C_k}(V_p) = 0 \quad \text{if } k > 2s. \tag{2.13}$$

Proof. The commutator maps of commuting creation operators commute. A factor whose support misses p can therefore be placed next to V_p and annihilates it. Expand a surviving nested commutator into products with some elementary operators to the left of V_p and the rest to its right. Outside p, the operator V_p changes nothing; every site created there must remain excited, and a site created twice makes that product zero. This proves (2.12). In a nonzero product the creation supports on each side of V_p are pairwise disjoint. Since each meets p, at most $|p| \leq s$ can occur on either side. More than $2s$ factors therefore give zero. □

Lemma 2.3 (Sector norm estimate)**.** *For $0 \le k \le 2s$ and creation operators $C_1, \dots, C_k$, define $\mathcal{B}_k(C_1, \dots, C_k)$ by its coefficients*

$$b_M = (H_0^M)^{-1} \Pi_M \operatorname{ad}_{C_1} \cdots \operatorname{ad}_{C_k}(V)\Omega, \qquad M \neq \varnothing. \tag{2.14}$$

For $k = 0$ the commutator product is the identity. Then

$$\|\mathcal{B}_k(C_1, \dots, C_k)\|_1 \le \frac{DJ}{\Delta} \beta_k(s) \prod_{i=1}^{k} \|C_i\|_1. \tag{2.15}$$

Proof. Expand each C_i into its elementary terms, and fix a tuple $(M_1, \dots, M_k, p)$. Denote its commutator by X as above. Its norm satisfies

$$\|X\Omega\| \le 2^k J \prod_{i=1}^{k} \|c_{M_i}^{(i)}\|. \tag{2.16}$$

Only tuples with $M_i \cap p \neq \varnothing$ need be retained. Fix a site u when estimating the norm in (2.10). We split the sum into tuples with $u \in p$ and those with $u \notin p$. In the latter case, a nonzero output sector containing u requires $u \in M_j$ for at least one j; summing over all such j is an upper bound.

If $u \in p$, put $n = |N_* \setminus p|$. A contributing output sector has the form (2.12) with $u \in T$ and $|M| = n + |T|$. Cauchy–Schwarz and (2.9) give

$$\begin{aligned} \sum_{M \ni u} \frac{\|\Pi_M X\Omega\|}{|M|} &\le \left(\sum_{\substack{T \subset p \\ u \in T}} \frac{1}{(n + |T|)^2} \right)^{1/2} \|X\Omega\| \\ &\le \gamma_0(s) \|X\Omega\|. \end{aligned} \tag{2.17}$$

For each fixed p and i,

$$\sum_{M_i : M_i \cap p \neq \varnothing} \|c_{M_i}^{(i)}\| \le \sum_{v \in p} \sum_{M_i \ni v} \|c_{M_i}^{(i)}\| \le s \|C_i\|_1. \tag{2.18}$$

There are at most D choices of p containing u. After using $\|(H_0^M)^{-1}\| \le 1/(\Delta|M|)$, the contribution of (2.17) is bounded by

$$\frac{DJ}{\Delta} 2^k \gamma_0(s) s^k \prod_i \|C_i\|_1.$$

Now suppose $u \notin p$ and $u \in M_j$. Then $n = |N_* \setminus p| \ge 1$ and $|M_j| \le n + s$. Hence

$$\begin{aligned} \sum_{M} \frac{\|\Pi_M X\Omega\|}{|M|} &\le \left(\sum_{t=0}^{s} \frac{\binom{s}{t}}{(n+t)^2} \right)^{1/2} \|X\Omega\| \\ &\le \frac{\gamma_1(s)}{|M_j|} \|X\Omega\|. \end{aligned} \tag{2.19}$$

For the last inequality, each function $(n+s)/(n+t)$ is nonincreasing for $n \ge 1$, since $t \le s$. Thus its square is bounded by $(s+1)^2/(t+1)^2$, and (2.4) applies. This proves the estimate for every n; no finite enumeration of sector sizes is needed.

For fixed M_j, at most $D|M_j|$ members p intersect M_j. This cancels the denominator $|M_j|$ in (2.19). Sum the other $k-1$ supports using (2.18), and then sum $M_j \ni u$. For this j the result is bounded by

$$\frac{DJ}{\Delta} 2^k \gamma_1(s) s^{k-1} \prod_i \|C_i\|_1.$$

Adding the k possible choices of j proves (2.15). For $k = 0$, only the first case occurs, with $N_* = \varnothing$, and its bound is $DJ\gamma_0(s)/\Delta$. □

Proof of Theorem 2.1 in finite dimension. For a creation operator C, put

$$W(C) = e^C V e^{-C} = \sum_{k=0}^{2s} \frac{1}{k!} \operatorname{ad}_C^k(V). \tag{2.20}$$

The equality is exact by Lemma 2.2. By (2.11), $e^{-C}\Omega$ is an eigenvector of $H(\varepsilon)$ precisely when

$$c_M = \varepsilon (H_0^M)^{-1} \Pi_M W(C)\Omega \quad (M \neq \varnothing). \tag{2.21}$$

Its eigenvalue is $E(\varepsilon) = \varepsilon\langle\Omega, W(C)\Omega\rangle$. Indeed, conjugating H by e^C gives $H_0 + [C, H_0] + \varepsilon W(C)$; its nonvacuum components on Ω vanish exactly when (2.21) holds.

Let F_ε denote the right side of (2.21), assembled as a creation operator. Lemma 2.3 gives

$$\|F_\varepsilon(C)\|_1 \leq \lambda q_s(\|C\|_1). \tag{2.22}$$

The same multilinear estimate, applied to the difference of two k-fold commutators, gives, whenever $\|C\|_1, \|C'\|_1 \leq r$,

$$\|F_\varepsilon(C) - F_\varepsilon(C')\|_1 \leq \lambda q_s'(r)\|C - C'\|_1. \tag{2.23}$$

For example, the difference of the k-th terms is a sum of k multilinear terms containing one $C - C'$ and $k - 1$ factors bounded by r. This explains the derivative of q_s in the estimate. By (2.7), the closed ball of radius r is preserved and the contraction factor is less than $1/2$. The Banach fixed-point theorem supplies a solution C with $\|C\|_1 \leq r$. In finite volume the space of creation coefficients is complete in (2.10). The vector $e^{-C}\Omega$ is nonzero, since its vacuum coefficient is one.

We next prove isolation and simplicity of this eigenvalue. Suppose $H\phi = (E+\delta)\phi$ with $|\delta| < \Delta/2$, where ϕ is linearly independent of $e^{-C}\Omega$. There is a unique creation operator B and scalar b_0 such that

$$e^C\phi = (B + b_0)\Omega.$$

Here $B \neq 0$. The nonvacuum components of the eigenvalue equation after conjugation yield

$$b_M = \varepsilon (H_0^M - \delta)^{-1} \Pi_M [B, W(C)]\Omega. \tag{2.24}$$

To verify the sign, note that $[H_0 + [C, H_0] + \varepsilon W(C), B] = [H_0, B] - \varepsilon[B, W(C)]$. The creation part $[C, H_0]$ commutes with B. Since $|M| \geq 1$,

$$\|(H_0^M - \delta)^{-1}\| \leq \frac{1}{\Delta|M| - |\delta|} \leq \frac{2}{\Delta|M|}. \tag{2.25}$$

The proof of Lemma 2.3 therefore applies with an extra factor two. In (2.20), commutation with B increases the order by one; orders greater than $2s$ vanish. Consequently

$$\begin{aligned} \|B\|_1 &\leq 2\lambda \sum_{k=0}^{2s-1} \frac{\beta_{k+1}(s)}{k!} \|C\|_1^k \|B\|_1 \\ &\leq 2\lambda q_s'(r)\|B\|_1 < \|B\|_1, \end{aligned} \tag{2.26}$$

a contradiction. This includes $\delta = 0$, so E is simple.

Finally vary the real parameter from 0 to ε. The same ball and contraction factor work throughout this segment, and (2.23) implies continuous dependence of the fixed point on the parameter. Thus E is a continuous simple eigenvalue branch, separated by at least $\Delta/2$ from every other eigenvalue. At zero it is the ground eigenvalue of H_0. Continuity of the finite-dimensional eigenvalues prevents a crossing, proving (2.8). □

Extension to compact-resolvent site operators. Let $P_{u,R} = \mathbf{1}_{[0,R]}(h_u)$ and $\mathcal{F}_R = \bigotimes_u \operatorname{Ran} P_{u,R}$. These are finite-dimensional spaces containing the product vacuum. Compress H to $\mathcal{F}_R$. Each onsite gap remains at least Δ, and each compressed interaction is supported on the same p and has norm at most J. The finite-dimensional theorem therefore applies with the same constants.

In a finite system V is bounded. Hence H is self-adjoint on $\operatorname{Dom} H_0$, bounded below, and has compact resolvent. The nested union of the spaces $\mathcal{F}_R$ contains all finite linear combinations of product eigenvectors of the h_u, and is an operator core, hence also a form core, for H_0 and H. The min–max principle gives, for each fixed eigenvalue index j,

$$E_j(H|_{\mathcal{F}_R}) \downarrow E_j(H),$$

where $H|_{\mathcal{F}_R}$ denotes compression and eigenvalues are ordered with multiplicity. If the full space has dimension at least two, pass to the limit in $E_1(H|_{\mathcal{F}_R}) - E_0(H|_{\mathcal{F}_R}) \geq \Delta/2$. The resulting positive difference proves both the asserted gap and simplicity. For a one-dimensional full space, the spectral exclusion is vacuous and the ground state is simple. No norm of an unbounded onsite operator is used in the estimates. □

2.2 Explicit rational certificates

The following choices make the criterion explicit:

$$\lambda_3 = 0.01790402, \qquad \lambda_4 = 0.01190829. \tag{2.27}$$

Every terminating decimal in this subsection denotes an exact rational number. To check the inequalities without locating a polynomial root, use the following rational upper bounds and witnesses:

s	$\overline{\gamma}_0$	$\overline{\gamma}_1$	r
3	1.269295517644	5.859465277083	0.03911425
4	1.464866319271	8.597480251020	0.02996979

The exact squares required by (2.4) are

s	$\gamma_0(s)^2$	$\gamma_1(s)^2$
3	29/18	103/3
4	103/48	887/12

. (2.28)

Squaring the displayed rational upper bounds verifies $\overline{\gamma}_i^2 \geq \gamma_i^2$. Let $\overline{q}_s$ be (2.6) with these upper bounds substituted in (2.5). Direct rational substitution gives, for each of the two rows,

$$r - \lambda_s \overline{q}_s(r) > \frac{3}{10^8}, \qquad 1 - 2\lambda_s \overline{q}_s'(r) > \frac{9}{10^7}. \tag{2.29}$$

These are finite rational inequalities, verifiable by clearing denominators. Since q_s and q_s' have nonnegative coefficients bounded by those of $\overline{q}_s$, they imply (2.7) for all $0 \leq \lambda \leq \lambda_s$. The certificate checks only this arithmetic; the universal operator conclusion comes from the preceding proof. It does not follow from finite matrix diagonalizations or from sampling interaction operators.

2.3 Application to the specified SU(N) lattice Hamiltonian

Let $d \geq 2$ be the spatial dimension, $N \geq 2$, $x = g^2 > 0$, and $0 \leq b_N \leq 2N$. On a finite collection E of positively oriented cubic-lattice links, retain any subset $\mathcal{P}$ of the usual elementary plaquettes whose four links belong to E, with no repetitions of an anchored plaquette. Reverse orientations denote inverses of the positive link variables. At lattice spacing one, use

$$H_{\mathrm{lat}}(x) = \frac{x}{2} \sum_{l \in E} C_l + \frac{b_N}{x} \sum_{p \in \mathcal{P}} (1 - T_p), \qquad T_p = \frac{\mathrm{Re}\,\mathrm{Tr}\, U_p}{N}. \tag{2.30}$$

The link space is $L^2(\mathrm{SU}(N), \mathrm{d}U)$ for normalized Haar measure; C_l is the positive group Casimir with $\mathrm{Tr}(t_a t_b) = \delta_{ab}/2$. The physical space is invariant under the vertex gauge groups, with no external charges. These conventions specify the magnetic normalization explicitly; compare the Hamiltonian formulation in [3]. The bounds below concern the dimensionless energies of (2.30).

The least nonzero Casimir is

$$C_F = \frac{N^2 - 1}{2N} \geq \frac{3}{4}. \tag{2.31}$$

For completeness, write a nonzero dominant highest weight as $\Lambda = \omega_i + \nu$, where ν is dominant and the roots have squared length two. The Gram entries $(\omega_i, \omega_j) = \min(i, j)(N - \max(i, j))/N$ are positive, so $C_2(\Lambda) \geq C_2(\omega_i)$ for $C_2(\Lambda) = (\Lambda, \Lambda + 2\rho)/2$, where $\rho = \sum_{j=1}^{N-1} \omega_j$. Since $C_2(\omega_i) = i(N - i)(N + 1)/(2N) \geq C_F$, and the fundamental representation attains equality, Peter–Weyl decomposition proves (2.31) on the entire link space.

Divide (2.30) by $s_0 = xC_F/2$ and remove its scalar magnetic constant. Precisely,

$$\frac{H_{\mathrm{lat}}}{s_0} - \varepsilon |\mathcal{P}| I = \sum_{l \in E} \frac{C_l}{C_F} + \varepsilon \sum_{p \in \mathcal{P}} (-T_p), \qquad \varepsilon = \frac{2b_N}{x^2 C_F} \leq \frac{32}{3x^2}. \tag{2.32}$$

The last inequality uses $2b_N/C_F \leq 8N^2/(N^2 - 1) \leq 32/3$. The normalized onsite gap is $\Delta = 1$, $J = 1$ because $\|T_p\| \leq 1$, and each plaquette meets $s = 4$ link sites. Each link belongs to at most $D = 2(d - 1)$ elementary plaquettes. These statements hold for any finite open subgraph, and also for the usual periodic directed cubic lattices of side lengths at least two, with positive-direction link factors distinguished.

Corollary 2.4 (Explicit N-uniform finite-volume gauge gap)**.** *For the model (2.30), let*

$$X_d^{\mathrm{link}} = \left(\frac{(32/3)\, 2(d - 1)}{0.01190829} \right)^{1/2}. \tag{2.33}$$

If $x \geq X_d^{\mathrm{link}}$, the ground state is unique and gauge invariant, and both on the full link space and on its physical subspace,

$$H_{\mathrm{lat}} - E_0 \geq \frac{xC_F}{4} \geq \frac{3x}{16} \quad \text{on the orthogonal complement of the ground state.} \tag{2.34}$$

In three spatial dimensions the sufficient condition $x \geq 59.9$ implies (2.34), for every $N \geq 2$ and $0 \leq b_N \leq 2N$, uniformly over the stated finite volumes.

Proof. Equations (2.32)–(2.33) give $\lambda = \varepsilon D \leq \lambda_4$. Theorem 2.1 and the certificate give a normalized gap at least $1/2$. Restoring s_0 gives $xC_F/4$; adding back the scalar constant changes no gap. All Casimirs and plaquette traces commute with the finite product of vertex gauge groups. The unique ground line therefore carries a continuous one-dimensional representation of a product of SU(N)

groups. Such a representation is trivial, so the ground vector is gauge invariant. The physical space reduces H_{lat} and contains its ground state; restricting the excited-space inequality cannot lower the gap. Finally, the exact rational inequality $(59.9)^2(0.01190829) > (32/3)\,4$ verifies the stated threshold. □

For reference, upward-rounded sufficient thresholds using link sites are 42.4, 59.9, and 73.4 in spatial dimensions 2, 3, and 4, respectively. The exact values from (2.33) are approximately 42.326, 59.858, and 73.310; these approximations are not used as downward-rounded sufficient bounds. Alternatively, group the d outgoing positive links at each vertex into one site. Each plaquette then meets three sites, and a vertex site meets at most $3\binom{d}{2}$ plaquettes: for each coordinate pair its three possible roles are the lower anchor and its two forward neighbors. With $s = 3$, $D = 3d(d-1)/2$, and λ_3 from (2.27), the sufficient threshold is $\big((32/3)D/\lambda_3\big)^{1/2}$. In $d = 2$, this permits the slightly smaller upward-rounded value 42.3. Link grouping gives the better displayed bound in $d = 3$.

Remark 2.5 (Limits of the conclusion). The uniformity established here is over finite volume and N at fixed x. The proof does not construct an infinite-volume GNS Hamiltonian at the explicit threshold, nor does it prove a continuum limit or a positive physical continuum mass. An infinite-volume strong-coupling construction with a non-explicit threshold is a separate consequence of the stability theorem in [2], as applied in [6]. A limit with fixed 't Hooft coupling Nx eventually leaves the present window as N increases.

2.4 A counting correction in the cited BDL preprint

This subsection concerns the exact version arXiv:0707.1894v1 of [1], Appendix A, p. 25, in the paragraph preceding its equation (69). The intermediate inference $M_j \subset M \cup \{w\}$, and hence $|M_j| \le 2|M|$, does not follow from the support conditions stated there. The following example also verifies that this is an operator issue, rather than only a set-theoretic possibility.

Take three distinct qubit sites u, v, w, their vacuum $|000\rangle$, and

$$A = a_u^\dagger a_v^\dagger a_w^\dagger, \qquad V_{vw} = X_v X_w, \qquad a^\dagger = |1\rangle\langle 0|, \quad a = |0\rangle\langle 1|.$$

Here X interchanges $|0\rangle$ and $|1\rangle$. Direct application gives

$$AV_{vw}|000\rangle = 0, \qquad V_{vw}A|000\rangle = |100\rangle, \qquad \langle 000|a_u[A, V_{vw}]|000\rangle = -1. \tag{2.35}$$

Thus a nonzero term has $M_j = \{u, v, w\}$ and $M = \{u\}$. It obeys the source's support conditions but violates $|M_j| \le 2|M|$.

The valid general estimate is

$$M_j \subset M \cup p, \quad |p| \le s, \quad M \neq \varnothing \quad \Longrightarrow \quad |M_j| \le |M| + s \le (s+1)|M|. \tag{2.36}$$

For two-body interactions this gives $3|M|$, which is sharp by (2.35). It does not invalidate the conservative constant 2^{13} in the source's Lemma 3. Repeating its counting with the factor three instead of two gives, for the possible commutator orders $1 \le k \le 4$, the coefficient bound

$$k \cdot 4 \cdot 3 \cdot 2^k 2^{k-1} + 4 \cdot 2^k 2^k \le 7168 < 8192 = 2^{13}. \tag{2.37}$$

Higher orders vanish, and the zeroth-order estimate is smaller. Accordingly the correction repairs this intermediate counting step while preserving that theorem's stated conservative constant. No priority claim is made for identifying the correction, and no claim about revisions other than the specified source version is needed here.

3 Fixed-block coercivity in the actual vacuum

This section proves a lower bound for the energy of fluctuations in a fixed finite set of links, conditional on all exterior links, in the actual finite-volume ground state. The constant is independent of the number of exterior links. Its dependence on the number of selected links is explicit and does not give a uniform bound for an extensive elimination layer. We use a Hamiltonian of Kogut–Susskind type [3] with the normalization specified below. The argument is presented in full; the associated research record is available in [7].

3.1 Finite-volume Hamiltonian and its ground-state form

Let E be a finite set of oriented link representatives of a finite spatial graph, let $\mathcal{P}$ be a finite collection of plaquette terms, and fix $N \geq 2$, $x > 0$, $a > 0$, and $b \geq 0$. Put $M = \mathrm{SU}(N)^E$ and equip M with normalized product Haar measure $\mathrm{d}U$. For Hermitian generators satisfying $\mathrm{Tr}(t_\alpha t_\beta) = \delta_{\alpha\beta}/2$, choose the bi-invariant metric for which the corresponding left-invariant vector fields X_α are orthonormal. On link ℓ, write $C_\ell = -\sum_\alpha X_{\ell,\alpha}^2 = -\Delta_\ell$. Consider

$$H = \frac{x}{2} \sum_{\ell \in E} C_\ell + V, \qquad V(U) = \frac{b}{x} \sum_{p \in \mathcal{P}} \left(1 - \frac{\mathrm{Re}\,\mathrm{Tr}\, U_p}{N} \right). \tag{3.1}$$

Here U_p is the ordered plaquette holonomy. Each summand in parentheses lies in $[0, 2]$. The operator H is the self-adjoint bounded-potential perturbation of the product Laplacian, with operator domain $H^2(M)$ and form domain $H^1(M)$. The parameter a enters the physical energy through division by a; it has not been absorbed into H.

The vertex gauge group acts by $U_\ell \mapsto g_{s(\ell)} U_\ell g_{t(\ell)}^{-1}$. We assume the plaquette terms and boundary conditions preserve this action. Physical functions are invariant under this group. No gauge fixing is used.

Lemma 3.1 (Actual vacuum and ground-state transform)**.** *The lowest eigenvalue E_0 of H is simple. Its normalized eigenfunction ψ can be chosen smooth, strictly positive, and gauge invariant. Set $\mathrm{d}\mu = \psi^2\, \mathrm{d}U$. Multiplication by ψ is a unitary map from $L^2(\mu)$ to $L^2(\mathrm{d}U)$, and the nonnegative self-adjoint operator*

$$K = \psi^{-1}(H - E_0)\psi/a \tag{3.2}$$

has closed quadratic form

$$k[f] = \frac{x}{2a} \int_M \sum_{\ell,\alpha} |X_{\ell,\alpha} f|^2\, \mathrm{d}\mu, \qquad f \in H^1(M). \tag{3.3}$$

Its kernel is the space of constant functions. These statements also hold after restriction to the gauge-invariant subspaces.

Proof. The compact manifold M is connected, whether or not the underlying graph is connected. Its heat kernel is strictly positive at every positive time. The Feynman–Kac weight for the bounded real potential V is strictly positive, so $\exp(-tH)$ is positivity improving. Compact resolvent gives a lowest eigenfunction. The absolute-value inequality for the Dirichlet form allows a nonnegative minimizer, which is strictly positive by the strong maximum principle. Positivity improvement implies simplicity: a second real ground eigenfunction orthogonal to the positive one would change sign, contradicting the strict positivity statement. Elliptic regularity gives smoothness. Gauge transformations commute with H and preserve positivity and norm; uniqueness therefore makes ψ gauge invariant.

For smooth complex f, integration by parts and $H\psi = E_0\psi$ give

$$\langle \psi f, (H - E_0)\psi f\rangle_{L^2(\mathrm{d}U)} = \frac{x}{2}\int_M \sum_{\ell,\alpha} |X_{\ell,\alpha} f|^2 \psi^2 \,\mathrm{d}U. \tag{3.4}$$

Indeed, the terms containing $|f|^2|\nabla\psi|^2$ and the mixed derivatives combine, after integration by parts, with $V|\psi f|^2$ to give $E_0|\psi f|^2$. On this fixed finite graph, ψ is smooth and bounded away from zero. Multiplication by ψ and its inverse preserve $H^1(M)$, so the identity extends by form closure. Unitarity follows directly from $\mathrm{d}\mu = \psi^2 \mathrm{d}U$. The zero-energy functions have zero gradient and are constant because M is connected. Multiplication by ψ intertwines the gauge actions, which proves the final assertion. □

In particular, all energy estimates below subtract E_0, the actual interacting ground energy. No bound on E_0 uniform in volume is assumed. The equivalence of weighted and unweighted Sobolev norms just used is needed only for each fixed graph; its constants need not be uniform in volume.

3.2 Conditional projection and finite-block theorem

Fix a nonempty subset $B \subseteq E$ with $m = |B|$, and write a configuration as $(u, y) \in \mathrm{SU}(N)^B \times \mathrm{SU}(N)^{E\setminus B}$. Let p_B be the number of plaquette terms that involve at least one link in B, counting each term once. Define

$$Z(y) = \int \psi(v, y)^2 \,\mathrm{d}v, \qquad \mathrm{d}\mu_y(u) = \frac{\psi(u, y)^2}{Z(y)}\,\mathrm{d}u, \qquad Pf(u, y) = \int f(v, y)\,\mathrm{d}\mu_y(v), \tag{3.5}$$

where $\mathrm{d}u$ and $\mathrm{d}v$ are normalized product Haar measures. Thus $P = \mathbb{E}_\mu[\cdot \mid y]$ is an orthogonal projection in $L^2(\mu)$ and $P1 = 1$. Put $Q = I - P$.

The restriction of the form k to $H^1(M) \cap \operatorname{Ran} Q$ is closed and densely defined in $\operatorname{Ran} Q$; the density assertion is verified in the proof below. Let D_B be its associated self-adjoint operator on that subspace. Thus D_B denotes a form compression. It does not require the unbounded operator product QKQ to be defined on an assumed common operator domain.

Write

$$C_F = \frac{N^2 - 1}{2N}, \qquad n = N^2 - 1, \qquad \kappa_N(s) = \frac{\max_{u,v\in \mathrm{SU}(N)} k_s(u, v)}{\min_{u,v\in \mathrm{SU}(N)} k_s(u, v)}, \quad s > 0, \tag{3.6}$$

where k_s is the one-link heat kernel of $\exp(-sC) = \exp(s\Delta)$ relative to normalized Haar measure. Compactness and strict positivity make $\kappa_N(s)$ finite.

Theorem 3.2 (Exterior-volume-uniform finite-block coercivity)**.** *For every $s > 0$,*

$$D_B \;\geq\; d_B(s) I, \qquad d_B(s) = \frac{xC_F}{2a}\,\kappa_N(s)^{-2m} \exp\!\left(-\frac{8sbp_B}{x^2}\right). \tag{3.7}$$

The bound is independent of the number of links outside B when m, p_B, N, x, b, a are fixed. The projection P commutes with the gauge action, so the same bound holds on the gauge-invariant part of $\operatorname{Ran} Q$*, if that part is nonzero. If $bp_B = 0$, the sharper bound*

$$D_B \geq \frac{xC_F}{2a} I \tag{3.8}$$

holds.

Proof. Split the potential into terms that avoid and meet B:

$$V(u,y) = V_{\text{out}}(y) + V_{\text{hit}}(u,y), \qquad 0 \le V_{\text{hit}} \le W_B := \frac{2bp_B}{x}. \tag{3.9}$$

Let (U_t, Y_t) be independent link Brownian motions, with generator $(x/2)\Delta$ on each link. At any time $t > 0$, the eigenfunction identity and Feynman–Kac formula give

$$e^{-tE_0}\psi(u,y) = \mathbb{E}_{u,y}\left[e^{-\int_0^t V(U_r,Y_r)\,\mathrm{d}r}\psi(U_t,Y_t)\right]. \tag{3.10}$$

Remove only V_{hit} from this expectation and call the result $F_t(u,y)$. Positivity gives

$$e^{-tW_B}F_t(u,y) \le e^{-tE_0}\psi(u,y) \le F_t(u,y). \tag{3.11}$$

Put $s = xt/2$. Independence of the link motions and the absence of u from V_{out} imply

$$\begin{aligned} F_t(u,y) &= \int k_s^B(u,v)A_t(y,v)\,\mathrm{d}v, \\ A_t(y,v) &= \mathbb{E}_y\left[e^{-\int_0^t V_{\text{out}}(Y_r)\,\mathrm{d}r}\psi(v,Y_t)\right] > 0, \end{aligned} \tag{3.12}$$

where k_s^B is the product of the m one-link kernels. For a fixed y, the positive factor $A_t(y,v)$ is the same for every initial block configuration u. Consequently, for every u, u', y,

$$\frac{\psi(u,y)}{\psi(u',y)} \le e^{tW_B}\frac{F_t(u,y)}{F_t(u',y)} \le e^{tW_B}\kappa_N(s)^m =: R_B(s). \tag{3.13}$$

The factor e^{-tE_0} cancels in this same-time ratio. The exterior potential remains inside the common positive expectation in (3.12); it has not been replaced by a bound depending on the number of exterior plaquettes. This is the source of the asserted exterior-volume independence.

We next recall the product Haar Poincaré inequality in the chosen metric:

$$\mathrm{Var}_{\mathrm{d}u}(f) \le \frac{1}{C_F}\int \sum_{\ell\in B,\alpha} |X_{\ell,\alpha}f|^2\,\mathrm{d}u. \tag{3.14}$$

For completeness, Peter–Weyl decomposition reduces this statement to the smallest nonzero Casimir eigenvalue of $\mathrm{SU}(N)$. With long roots of squared length 2, the Casimir of highest weight λ is $(\lambda, \lambda + 2\rho)/2$. The fundamental weight ω_j has Casimir $j(N-j)(N+1)/(2N)$. All pairings of fundamental weights are nonnegative, so every nonzero dominant integral weight has Casimir at least the minimum of these fundamental values, namely C_F. On a product, the eigenvalues add; a nonconstant component therefore has eigenvalue at least C_F. This proves (3.14), including for complex functions by applying the real inequality to their real and imaginary parts.

For any positive probability density w relative to product Haar with $\max w / \min w \le R^2$, the minimizing-constant definition of variance and (3.14) give

$$\begin{aligned} \mathrm{Var}_{w\mathrm{d}u}(f) &\le (\max w)\,\mathrm{Var}_{\mathrm{d}u}(f) \\ &\le \frac{\max w}{C_F}\int |\nabla_B f|^2\,\mathrm{d}u \le \frac{R^2}{C_F}\int |\nabla_B f|^2 w\,\mathrm{d}u, \end{aligned} \tag{3.15}$$

where $|\nabla_B f|^2 = \sum_{\ell\in B,\alpha}|X_{\ell,\alpha}f|^2$. No Haar-mean-zero condition is imposed on f in this comparison. By (3.13), the density of μ_y in (3.5) has maximum-to-minimum ratio at most $R_B(s)^2$. Applying (3.15) conditionally and integrating over the exterior marginal yields

$$\|Qf\|^2_{L^2(\mu)} \le \frac{R_B(s)^2}{C_F}\int_M |\nabla_B f|^2\,\mathrm{d}\mu. \tag{3.16}$$

If $f \in \operatorname{Ran} Q$, the exact ground-state form therefore gives

$$k[f] \geq \frac{x}{2a} \int_M |\nabla_B f|^2 \, \mathrm{d}\mu \geq \frac{x C_F}{2a R_B(s)^2} \|f\|^2_{L^2(\mu)}. \tag{3.17}$$

Since $t = 2s/x$ and $W_B = 2bp_B/x$, this is precisely (3.7).

To justify the form restriction, observe that P maps smooth functions to smooth functions. On each fixed compact graph it is bounded on $H^1(M)$: differentiating its conditional integral in an exterior direction produces the conditional derivative of f and a covariance with $2\nabla_y \log \psi$. These coefficients are smooth and bounded, and conditional Cauchy–Schwarz gives the required fixed-graph bound. Derivatives in the B directions vanish for Pf. Thus Q is also bounded on $H^1(M)$. If $f \in H^1(M) \cap \operatorname{Ran} Q$, approximate f in H^1 by smooth f_j and use $Qf_j \to f$. This proves smooth density in the restricted form domain. Smooth density in L^2 and boundedness of Q also prove that the restriction is densely defined in $\operatorname{Ran} Q$. It is closed because this subspace is closed in $L^2(\mu)$ and k is closed. Inequality (3.17) consequently extends to the entire restricted form domain without any uniform derivative bound on ψ.

For a fixed gauge transformation, the selected and exterior link coordinates transform separately. Both transformations preserve their Haar measures, and ψ is invariant. Changing variables in (3.5) therefore shows that P intertwines the gauge action. The forms and their restrictions are gauge invariant, proving the physical-sector assertion.

Finally, if $bp_B = 0$, the Hamiltonian splits as a free product Laplacian on B plus an exterior operator. Its positive ground function is constant on B. The conditional measures are exactly product Haar, and (3.14) gives (3.8). □

3.3 An explicit estimate at weak coupling

Let D_N be the diameter of $\mathrm{SU}(N)$ in the specified Casimir metric. The next heat-kernel estimate supplies explicit constants in Theorem 3.2.

Lemma 3.3 (Compact heat-kernel ratio)**.** *For every $s > 0$,*

$$\kappa_N(s) \leq 2^n \exp\left(\frac{3D_N^2}{4s}\right), \qquad n = N^2 - 1. \tag{3.18}$$

Proof. The bi-invariant metric has nonnegative Ricci curvature. For a positive solution of $u_t = \Delta u$ on a compact n-dimensional manifold with nonnegative Ricci curvature, the Li–Yau estimate [4] is

$$|\nabla \log u|^2 - \partial_t \log u \leq \frac{n}{2t}. \tag{3.19}$$

Here is its compact, boundaryless derivation. Set $f = \log u$, $w = |\nabla f|^2 - f_t$, and $\mathcal{F} = tw$. The heat equation, Bochner identity, and $|\operatorname{Hess} f|^2 \geq (\Delta f)^2/n$ imply

$$(\Delta - \partial_t)\mathcal{F} \geq \frac{2\mathcal{F}^2}{nt} - \frac{\mathcal{F}}{t} - 2\langle \nabla f, \nabla \mathcal{F} \rangle. \tag{3.20}$$

For smooth positive initial data, $\mathcal{F} = 0$ at $t = 0$. A positive space-time maximum exceeding $n/2$ contradicts (3.20) and the maximum principle. This proves (3.19). For the heat kernel, start after a positive time shift and then let that shift tend to zero. Compactness removes boundary and escape issues.

Along a constant-speed minimizing geodesic γ between times $0 < t_1 < t_2$, inequality (3.19) gives

$$\frac{\mathrm{d}}{\mathrm{d}t} f(\gamma(t), t) \geq -\frac{|\dot{\gamma}(t)|^2}{4} - \frac{n}{2t}.$$

Integration yields

$$u(z, t_1) \leq u(w, t_2) \left(\frac{t_2}{t_1}\right)^{n/2} \exp\left(\frac{\mathrm{dist}(z, w)^2}{4(t_2 - t_1)}\right). \tag{3.21}$$

Apply this to $k_t(v, \cdot)$ between s and $2s$ and integrate the later point w against normalized Haar measure. Total heat-kernel mass one gives

$$\max k_s \leq 2^{n/2} e^{D_N^2/(4s)}. \tag{3.22}$$

Apply (3.21) between $s/2$ and s and integrate the earlier point z instead. The result is

$$\min k_s \geq 2^{-n/2} e^{-D_N^2/(2s)}. \tag{3.23}$$

The two inequalities prove (3.18). Normalizing Riemannian volume to Haar probability multiplies the heat kernel by a constant and does not change the logarithmic differential estimate; its mass in the normalized measure is one, as used above. □

Corollary 3.4 (Explicit finite-block lower bound)**.** *If $bp_B > 0$, then*

$$D_B \geq \frac{xC_F}{2a}\, 2^{-2mn} e^{-C_B/x} I, \qquad C_B = 4D_N \sqrt{3mbp_B}. \tag{3.24}$$

Proof. Substituting (3.18) into (3.7) gives the lower certificate

$$\frac{xC_F}{2a}\, 2^{-2mn} \exp\left(-\frac{3mD_N^2}{2s} - \frac{8sbp_B}{x^2}\right).$$

The positive exponent is minimized at

$$s_* = \frac{xD_N}{4}\sqrt{\frac{3m}{bp_B}}, \qquad \frac{3mD_N^2}{2s_*} + \frac{8s_* bp_B}{x^2} = \frac{4D_N\sqrt{3mbp_B}}{x}.$$

This proves (3.24). The case $bp_B = 0$ is already covered by (3.8) and does not require this optimization. □

The exponential dependence is a rigorous lower certificate, not an asymptotic formula for the actual discarded spectrum as $x \downarrow 0$. In particular, the estimate is not uniform in m or N.

3.4 Conditional scale budget and the remaining limitation

One can quantify the strength of this certificate without asserting a renormalization theorem. Suppose, as an additional hypothesis, that the same fixed-block estimate with constants $c_0 = (C_F/2)2^{-2mn}$ and $C = C_B$ were available for every relevant exact coarse operator. Consider the illustrative scale and coupling sequences

$$a_r = AL^{-r}, \qquad x_r = (\sigma + qr)^{-1}, \qquad r \geq 1, \quad A > 0,\ L > 1,\ \sigma > 0,\ q \geq 0. \tag{3.25}$$

The reciprocal costs of the certified bounds $d_r^{\mathrm{cert}} = c_0 x_r e^{-C/x_r}/a_r$ then satisfy the exact identity

$$\sum_{r\geq 1} \frac{1}{d_r^{\mathrm{cert}}} = \frac{A}{c_0} e^{C\sigma} \sum_{r\geq 1} (\sigma + qr)\rho^r, \qquad \rho = \frac{e^{Cq}}{L}. \tag{3.26}$$

This series is finite exactly when $\rho < 1$, in which case it equals

$$\frac{A}{c_0} e^{C\sigma} \left(\frac{\sigma\rho}{1-\rho} + \frac{q\rho}{(1-\rho)^2} \right). \tag{3.27}$$

No actual Yang–Mills coupling trajectory, choice of scale factor, or stability of the finite-block theorem under exact elimination is proved by this calculation. If $\rho \geq 1$, this particular certificate does not give a finite upper bound on the inverse-gap cost; the true discarded gaps could still give a finite sum.

Theorem 3.2 concerns one fixed finite block in the original Hamiltonian (3.1). A whole elimination layer generally contains an extensive number of links. If that layer is used as B, the factor 2^{-2mn} deteriorates exponentially in m, and C_B grows proportionally to m when p_B does. Eliminating bounded blocks one after another instead requires estimates for the successive exact coarse operators and control of the number of elimination steps. Neither follows from the theorem: the generated operators have not been shown to retain the form (3.1), and conditional bounds for separate blocks do not by themselves provide a collective conditional Poincaré inequality in an interacting vacuum. Thus the proved exterior-volume uniformity leaves open a whole-layer estimate for the actual compact-link ground measure that persists under exact coarse graining.

4 Exact elimination and its induced inner product

The estimates in the preceding sections concern genuine finite-lattice operators. To use such estimates along a sequence of scales one also needs a comparison between the fine and coarse excitation problems. This section proves an abstract comparison; identifying a Yang–Mills renormalization scheme satisfying all its hypotheses is a separate problem.

4.1 A one-step comparison

Let $\mathcal{H} = P \oplus Q$ be a finite-dimensional complex Hilbert space. Suppose $H \geq 0$ has nonzero kernel and block decomposition

$$H = \begin{pmatrix} A & B^* \\ B & D \end{pmatrix}, \qquad D \geq dI_Q, \quad d > 0. \tag{4.1}$$

Energies have already been shifted by the *actual* ground energy. Define

$$C = D^{-1}B, \quad S = A - B^*D^{-1}B, \quad Jp = (p, -Cp), \quad G = J^*J = I + C^*C, \quad K = G^{-1/2}SG^{-1/2}. \tag{4.2}$$

S is a quadratic energy on the retained coordinates, while G is the squared norm of their energy-minimizing graph lift. The normalized map $U = JG^{-1/2}$ is an isometry.

Proposition 4.1 (Metric-normalized Schur gap comparison)**.** *If* $\operatorname{spec} K \subset \{0\} \cup [\gamma, \infty)$ *for* $\gamma > 0$*, then*

$$H \geq \left(\gamma^{-1} + d^{-1}\right)^{-1} (I - \Pi_{\ker H}). \tag{4.3}$$

The kernels of K *and* H *have the same dimension. If* K *has a nonzero excited sector, then also* $\operatorname{gap} H \leq \operatorname{gap} K$*.*

Proof. For $r = q + Cp$, completion of squares gives

$$\langle (p,q), H(p,q)\rangle = \langle p, Sp\rangle + \langle r, Dr\rangle. \tag{4.4}$$

Both summands are nonnegative, and hence $\ker H = J \ker S$. Decompose $p = p_0 + p_\perp$ orthogonally in the G inner product, with $p_0 \in \ker S$. Since $\ker K = G^{1/2} \ker S$, $Jp_\perp \perp \ker H$ and

$$\|Jp_\perp\|^2 \le \gamma^{-1} \langle p, Sp\rangle.$$

The decomposition $(p,q) = Jp_0 + Jp_\perp + (0,r)$ therefore implies

$$\begin{aligned} \operatorname{dist}((p,q), \ker H) &\le \|Jp_\perp\| + \|r\| \\ &\le \gamma^{-1/2} \langle p, Sp\rangle^{1/2} + d^{-1/2} \langle r, Dr\rangle^{1/2}. \end{aligned}$$

Squaring and applying scalar Cauchy–Schwarz proves (4.3). The isometry U identifies $\ker K$ with $\ker H$. For the upper comparison, minimize the fine Rayleigh quotient over $U((\ker K)^\perp)$; on that subspace it equals the coarse Rayleigh quotient. □

Remark 4.2 (Closed-form extension). The same proof applies to a closed nonnegative form on $P \oplus Q$ when $D \ge dI$ is self-adjoint, $C : P \to Q$ is bounded, s is a closed nonnegative form, and the fine form has exactly the representation

$$h[(p,q)] = s[p] + \|D^{1/2}(q + Cp)\|^2$$

on the domain

$$\{(p,q) : p \in \operatorname{Dom} s,\ q + Cp \in \operatorname{Dom} D^{1/2}\}.$$

Here $\operatorname{Dom} s$ denotes the form domain. The coarse operator is the self-adjoint operator associated with $k[z] = s[G^{-1/2} z]$ on $G^{1/2} \operatorname{Dom} s$, not an unchecked product of unbounded operators. The distance argument extends on this domain. An application to an unbounded lattice Hamiltonian must verify these conditions; a finite-matrix computation does not verify them. No extension to unbounded C is asserted.

4.2 Why the metric matters

For $M \in \mathbb{R}$ consider

$$H_M = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 + M^2 & M \\ 0 & M & 1 \end{pmatrix}, \qquad P = \operatorname{span}\{e_1, e_2\}, \quad Q = \operatorname{span}\{e_3\}. \tag{4.5}$$

Then $D = 1$, $S = \operatorname{diag}(0,1)$ and $G = \operatorname{diag}(1, 1 + M^2)$, whereas

$$\operatorname{gap} H_M = \frac{M^2 + 2 - \sqrt{M^4 + 4M^2}}{2} \longrightarrow 0 \quad \text{as } |M| \longrightarrow \infty. \tag{4.6}$$

Thus positive lower bounds for D and the *unweighted* Schur operator S give no positive uniform lower bound for the fine gap. The correct coarse gap is $(1 + M^2)^{-1}$, and Proposition 4.1 yields the consistent lower bound $(M^2 + 2)^{-1}$.

4.3 An additive inverse-gap budget

Corollary 4.3 (Conditional multiscale bound)**.** *Suppose $H_0, \dots, H_k$ form a sequence of exact metric-normalized Schur reductions satisfying Proposition 4.1, or its closed-form extension, in the same physical energy units. At step j assume the discarded block satisfies $D_j \geq d_j I$ with $d_j > 0$. If H_k has gap at least $m_* > 0$, then*

$$\operatorname{gap} H_0 \geq \left(m_*^{-1} + \sum_{j=0}^{k-1} d_j^{-1} \right)^{-1}. \tag{4.7}$$

Proof. Iterate (4.3) backwards from the terminal operator. □

This avoids a fixed multiplicative loss per scale, but every operator and metric in the hypothesis must be the actual one generated by elimination. For example, assume *as a model profile* $a_r = AL^{-r}$, $x_r = (\sigma + qr)^{-1}$ for $r \geq 1$, where $A, \sigma > 0$, $L > 1$ and $q \geq 0$. If one could prove $d_r \geq c x_r / a_r$ with one $c > 0$, then

$$\sum_{r \geq 1} d_r^{-1} \leq \frac{A}{c} \left(\frac{\sigma}{L-1} + \frac{qL}{(L-1)^2} \right). \tag{4.8}$$

The geometric summation is exact. Neither the assumed trajectory nor the nonlinear discarded-sector estimate is established here.

4.4 Quantum Schur elimination need not preserve quadratic form

A separate obstruction is that the normalized reduction above is not the same operation as integrating a Gaussian coordinate density. Let $H = \mathrm{d}\Gamma(h)$ be the bosonic second quantization of

$$h = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Retain the subspace spanned by all $|n, 0\rangle$, so that the retained space contains the true Fock vacuum. Total particle number is preserved by H and by all the reduction maps. In the one-particle sector,

$$S_1 = \frac{3}{2}, \qquad G_1 = \frac{5}{4}, \qquad K_1 = \frac{6}{5}.$$

In the two-particle sector the fine matrix is

$$H_2 = \begin{pmatrix} 4 & \sqrt{2} & 0 \\ \sqrt{2} & 4 & \sqrt{2} \\ 0 & \sqrt{2} & 4 \end{pmatrix}$$

in the basis $|2, 0\rangle, |1, 1\rangle, |0, 2\rangle$. Direct inversion gives

$$S_2 = \frac{24}{7}, \qquad G_2 = \frac{58}{49}, \qquad K_2 = \frac{84}{29} \neq 2K_1.$$

Since $h \geq I$, the normalized graph restriction in the n-particle sector is bounded below by nI. Consequently K_1 and $K_2 < 3$ are indeed the first two positive levels. They are not equally spaced. The full Fock-space form construction is legitimate: on the discarded part of the n-particle sector, $D_n \geq nI$ and $\|B_n\| = \sqrt{n}$, so $\|D_n^{-1} B_n\| \leq n^{-1/2}$ and the graph operator is bounded. This example rules out automatic closure of the oscillator class under this quantum elimination. It does not rule out controlling the generated operators in a larger class.

5 The remaining Yang–Mills obligations

The results above do not establish the hypotheses of a continuum mass-gap theorem. The following are separate missing implications.

1. **Collective nonlinear coercivity.** The finite-block bound must be replaced or supplemented by an estimate for an entire elimination layer in the actual compact-link vacuum, uniform in volume and ultraviolet cutoff. Its validity for each fixed block does not imply such a collective bound.

2. **Control after exact elimination.** The estimates must survive the generated interactions, the new vacuum, the induced coarse metric and the appropriate form domains. The reduced operator has not been identified with another Hamiltonian of the bare Wilson form (1.2). Substitution of a running coupling into the strong-coupling corollary is therefore insufficient.

3. **A physical terminal gap.** At a fixed physical coarse scale, one must control the actual terminal operator and its metric. The explicit bound at large bare g^2 provides a possible comparison target, but no comparison from the actual renormalized operator to that target is proved here.

4. **A nontrivial continuum field theory and spectral transfer.** One must construct compatible renormalized observables, their limiting correlations and a field theory with the requisite positivity, locality, covariance and Yang–Mills short-distance behavior. A positive finite physical gap requires an estimate uniform along the relevant limit. There must also be a nonzero finite-energy sector: exclusion of spectrum from an interval in a vacuum-only Hilbert space would not solve the problem posed in [5].

The scale distinction is especially important. The strong-coupling bound is in lattice units. At a fixed admissible x, its physical lower bound is proportional to x/a and diverges as $a \downarrow 0$. Along a weak bare-coupling continuum trajectory, $x(a)$ instead tends to zero, outside the theorem's certified window. Neither observation disproves a continuum mass gap; both prevent an inference of such a gap from the fixed-coupling estimate.

There is also a concrete obstruction to treating every configuration as a positive quadratic fluctuation problem. On an even periodic cubic lattice in spatial dimension at least two, choose central SU(2) links

$$U_i(z) = (-I)^{\sum_{k<i} z_k}.$$

Every elementary plaquette is $-I$. For a real link field v, vary $U_i(z;t) = U_i(z)\exp(itv_i(z)\sigma_3/2)$. Then

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}\frac{b}{x}\sum_p \left(1 - \frac{\operatorname{Re}\operatorname{Tr} U_p(t)}{2}\right)\bigg|_{t=0} = -\frac{b}{4x}\sum_p |(\mathrm{d}v)_p|^2. \tag{5.1}$$

A circulation around one square has zero divergence, zero block mean in each link component and nonzero curl. At this central background the adjoint transports are the identity, so these are also the covariant conditions. It is a discarded direction for the corresponding componentwise block average and has strictly negative magnetic Hessian for $b > 0$. Thus an all-background positive-Hessian comparison cannot provide the required nonlinear estimate. A negative Hessian of the potential does not imply a negative excitation Hamiltonian or a small quantum gap. Large-field regions require their own estimates.

The infinite-volume issue is kept distinct from uniformity in finite volume. The general stability theorem proved here supplies a common finite-volume lower bound. We do not claim in this paper

to construct the infinite-volume state and its GNS Hamiltonian at the explicit threshold 59.9. The earlier application of Yarotsky's results addresses that construction in a sufficiently strong, nonexplicit window [2, 6].

6 Evidence, reproducibility and attribution

The supplementary source archive contains the full LaTeX project, a standard-library Python verifier, exact arithmetic results, selected historical proof notes, and selected Lean source certificates with their toolchain manifest. The principal analytic proofs are in this paper. The machine checks have the following narrower scope:

- rational upper bounds for the square-root sector constants, the majorant supersolution inequalities and safe coupling thresholds;
- the finite counting correction and its three-qubit matrix element;
- exact finite-dimensional Schur and oscillator arithmetic.

The Python verifier uses rational arithmetic for its acceptance conditions; decimal displays are explanatory only. No random search, floating-point eigensolver, or finite sample is used to infer an all-volume theorem. The optional Lean certificates concern scalar inequalities, finite sets and a finite matrix calculation. They do not formalize the analytic stability theorem, the Feynman–Kac construction, Li–Yau theory or continuum quantum field theory.

The archive's `README.md` gives build and verification commands, and `MANIFEST.json` records the included files and their SHA-256 hashes. Historical notes are included for provenance and may retain their original notation or approximate threshold displays. The statements and safe sufficient thresholds in the present manuscript govern this release. In particular, 73.4, rather than a downward rounded 73.3, is a safe decimal upper rounding of the displayed four-spatial-dimensional link threshold when that comparison is used. No claim of global novelty is inferred from the absence of an identified prior formula.

AI assistance and review status

OpenAI Codex assisted with source inspection, drafting, mathematical consistency checks, exact verification code, typesetting and release preparation. Separate AI-assisted readings were used to check the mathematical arguments and the scope of the claims. This is not independent human peer review, and it does not imply that every derivation has been manually checked by the named author. The release includes a review record stating the performed checks and their limitations. No AI system is listed as an author.

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References

[1] S. Bravyi, D. P. DiVincenzo and D. Loss, *Polynomial-time algorithm for simulation of weakly interacting quantum spin systems*, Commun. Math. Phys. **284** (2008), 481–507. doi:10.1007/s00220-008-0574-6. The counting assertion examined here is in Appendix A, printed p. 25 of arXiv:0707.1894v1.

[2] D. A. Yarotsky, *Quasi-particles in weak perturbations of non-interacting quantum lattice systems*, arXiv:math-ph/0411042v1 (2004).

[3] J. Kogut and L. Susskind, *Hamiltonian formulation of Wilson's lattice gauge theories*, Phys. Rev. D **11** (1975), 395–408. doi:10.1103/PhysRevD.11.395.

[4] P. Li and S.-T. Yau, *On the parabolic kernel of the Schrödinger operator*, Acta Math. **156** (1986), 153–201, Theorem 1.1. doi:10.1007/BF02399203.

[5] A. Jaffe and E. Witten, *Quantum Yang–Mills theory*, Clay Mathematics Institute problem description, Section 4. `https://www.claymath.org/wp-content/uploads/2022/06/yangmills.pdf`.

[6] C. P. Zimmerman, *From Scalar Certificates to Spectral Bounds: Radial Pulsations, Discrete Hardy Forms, and SU(N) Lattice Gauge Theory*, version 1.0, September 22, 2026. doi:10.5281/zenodo.22895866.

[7] C. P. Zimmerman, *The Zimmerman Programme*, software and research archive, version v9.0.0, October 5, 2026. doi:10.5281/zenodo.23169118.